

TDK-thesis

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Multi-Level Interpretability of VPNet for ECG Classification Using SHAP and GPT Explanations

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Abstract

Interpretability is critical for deploying artificial intelligence (AI) models in safety-critical applications, particularly for tasks involving biomedical signals such as electrocardiograms (ECGs). This paper presents an explainable AI framework for ECG classification that combines deep learning models with feature attribution and language-based interpretation. We evaluate two architectures, VPNet, which encodes ECG signals into compact variable projection (VP) representations, and a baseline convolutional neural network (CNN), and compare their internal reasoning using SHAP (SHapley Additive exPlanations). SHAP values are computed at multiple levels, including raw inputs and intermediate representations, revealing distinct attribution behaviors between models. To enhance clinical interpretability, we introduce a post-hoc explanation layer using OpenAI’s GPT-4o, which translates SHAP outputs into structured, human-readable clinical narratives. Results show that VPNet produces sparser and more interpretable SHAP profiles, while CNN explanations are more diffuse and less clinically intuitive. Additionally, SHAP-based correlation analysis reveals little agreement between the models, underscoring the role of architecture in shaping explainability. The proposed approach demonstrates a pathway for integrating explainable deep learning with language-based medical reasoning, contributing toward more transparent and trustworthy AI systems in healthcare.

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Chapter 1

Introduction

Deep learning has shown advanced development in the field of electrodiagram (ECG) classification, allowing accurate detection of various cardiac abnormalities [1, 2]. Despite these advances, clinical adoption of such models still remains limited due to their "black-box" nature. In the context of high risk safety-critical applications, such as medical decisions, interpretability is a necessity. Clinicians must understand the rationale and reasoning behind an AI model's predictions to make sure diagnostic reliability that complies with regulatory standards, and safeguard patient outcomes.

This gap between strong technical performance and limited clinical use highlights a key issue where although deep neural networks excel at detecting patterns in biomedical signals, they typically do not provide insight into how their decisions are made. This lack of transparency is particularly concerning in ECG analysis, where clinical interpretation relies heavily on understanding the shape and timing of specific waveforms, such as the P-wave, QRS complex, and T-wave [3]. If a model cannot explain its reasoning in terms that align with clinical practice, it is unlikely to gain trust or acceptance—no matter how accurate it is.

Recent developments in explainable artificial intelligence (XAI) have sought to address this issue. SHAP (SHapley Additive exPlanations), in particular, provides a mathematically grounded method for attributing importance to individual input features. While SHAP offers useful insights into model behavior, its outputs are often difficult for non-technical users, such as clinicians, to interpret and understand in a meaningful way. In parallel, large language models (LLMs) now offer the potential to translate these technical attributions into human-readable, clinically contextualized explanations.

In this work, we present a novel interpretability framework that integrates SHAP with

GPT-based textual explanations to improve the transparency of ECG classification models. We focus on three neural network architectures: VPNet (Variable Projection Network, a 1D convolutional neural network (CNN), and a fully connected neural network (FCNN). VPNet distinguishes itself by incorporating Hermite function decompositions, a form of domain knowledge encoding, which may lend itself to greater interpretability compared to standard architectures.

Our contributions are threefold. First, we conduct SHAP analysis not only on the model output level, but also at intermediate layers, including VP layers and fully connected layers, providing a deeper view of how each component contributes to classification. Second, we introduce a time-segment-to-wave-region mapping that aligns model attributions with physiologically meaningful regions of the ECG waveform. This mapping ensures that SHAP explanations can be interpreted in a clinically coherent way. Third, we use GPT-based language models, guided by American Heart Association terminology and clinical evidence, to generate textual explanations that prevent model hallucination in generating outputs and contextualize model behavior within cardiological reasoning.

Together, these components form a multi-level interpretability pipeline that offers both technical transparency and clinical relevance. Our findings offer a pathway toward building transparent and explainable AI systems for ECG interpretation, which may facilitate their integration into clinical workflows and enhance patient care.

Chapter 2

Related Work

2.1 Interpretability Techniques in Deep Learning for Healthcare

Interpretability has become a main concern in the application of machine learning to healthcare. A wide range of techniques has emerged to meet this need, from inherently interpretable models to post-hoc explanation methods designed for more complex, black-box systems [4]. In the clinical context, interpretability is not just desirable but essential. As Ghassemi et al. [5] argue, clinical development requires both accuracy and transparency to earn the trust of medical practitioners and support informed decision-making.

Among post-hoc methods, SHAP has gained recognition as a robust, model-agnostic approach grounded in cooperative game theory. Originally proposed by Lundberg and Lee [6], SHAP assigns each input feature a contribution value that quantifies its role in the model's prediction. Compared to simpler techniques, SHAP offers desirable theoretical guarantees, such as local accuracy, missingness (features not used by the model receive zero attribution), and consistency, which have made it particularly suitable for high-stakes domains. In healthcare, SHAP has been successfully applied to explain predictions in both medical imaging and biosignal analysis, including studies by Sayres et al. [7] on retinal disease detection and by Kwon et al. [8] on ECG classification.

Other interpretability methods include Lime [9], which builds locally linear surrogate models to approximate complex decision boundaries, and counterfactual explanations [10], which simulate hypothetical scenarios to reveal how predictions could change with minimal input alterations. These approaches are valuable for model auditing and de-

bugging, especially in clinical settings where "what-if" reasoning aligns with physicians' diagnostic thinking. However, most of these methods solely on input-output relationships, offering little insight into the model's internal feature transformations or hierarchical representations.

2.2 ECG-Specific Model Interpretability

ECG classification presents unique interpretability challenges due to the temporal dynamics of cardiac signals and the clinical importance of specific waveform components. Previous work has used saliency-based methods to identify signal segments influential to model predictions. For instance, Strodtz et al. [11] employed saliency maps to visualize which ECG regions most influenced CNN-based myocardial infarction predictions. Similarly, Raghunath et al. [12] used Grad-CAM to highlight activation patterns in arrhythmia classification, revealing alignment with known clinical markers.

Attention-based models have also shown promise in this domain. Hong et al. [13] demonstrated that transformers attention weights often focus on clinically relevant ECG segments, suggesting a degree of internal reasoning aligned with expert interpretation. However, as Sui et al. [14] caution, attention weights do not always reflect true model importance and may mislead users when interpreted as explanatory evidence.

A common limitation of these approaches is their narrow focus on the relationship between raw ECG inputs and final outputs. Few studies investigate how intermediate representations, such as learned filters, projections, or hidden activations, contribute to a model's decision. Moreover, little work has been done to bridge the gap between quantitative attribution methods and clinically meaningful language, limiting the practical utility of explanations in real world diagnostic workflows [15].

2.3 Language Models for Clinical Explanation

Recent progress in LLMs has opened a new possibilities for translating technical model outputs into accessible natural language explanations [16] [17] [18]. Early studies by Alsentz et al. [19] showed that domain specific pretraining significantly improves LLM performance on medical tasks. Building on this, Yang et al. [20] demonstrated that GPT-4V achieves high accuracy in interpreting multimodal clinical exam questions.

In the context of ECG analysis, recent studies have demonstrated that large language models, such as GPT-4, can generate narrative reports describing cardiac abnormalities based on model outputs. For instance, GPT-4’s interpretative accuracy improved significantly with the addition of clinical context, highlighting its potential to act as an explanatory interface aligned with expert cardiologist assessments [21]. This aligns with the framing by Bommasani et al. [22], who describe foundation models as central, adaptable systems capable of supporting a wide range of downstream applications, including translation of technical outputs into accessible language.

However, these explanations risk diverging from the model’s actual reasoning if not carefully constrained. To address this, Fayyaz et al. [23] proposed grounding natural language outputs in attribution-based evidence. Inspired by this approach, this work integrates GPT-based explanations in SHAP-derived feature importance values, ensuring that generated narratives reflect what the model truly considered significant.

2.4 Wavelet and Projection-Based Methods for ECG Modeling

ECG signals are highly structured and present characteristic morphologies that lend themselves well to basis function decompositions [24]. Hermite functions, in particular, have proven effective for ECG modeling, offering a compact and interpretable representation of the cardiac cycle. Studies by Sornmo et al. [25], Lagerholm et al. [26] and Sandryhaila et al. [27] demonstrate that just a few Hermite coefficients can capture essential waveform features such as peaks, widths, and asymmetries.

The Variable Projection (VP) framework introduced by Golub and Pereyra [28] provides a principled way to separate linear and nonlinear parameters in such decompositions. More recently, Kovacs et al. [29] introduced VPNet, a deep learning architecture [30] that integrates VP as a layer within neural networks. This allows the model to learn data-adaptive Hermite decompositions, effectively embedding domain knowledge into the architecture.

Unlike CNNs that learn arbitrary filters, VPNet produces features with inherent physiological meaning. This nonlinear components (translation and dilation) capture temporal alignment and spread, while the linear coefficients represent the amplitude and shape of waveform components. This structure not only enhances model interpretability but also

aligns well with clinical reasoning, making VPNet particularly suitable for sensitive applications like ECG analysis.

VPNet is built with interpretability in mind, especially in its VP layer, which maps ECG signals onto Hermite basis functions that have clear physiological meaning. This layer produces both coefficients and nonlinear parameters, such as dilation and translation, offering insight into how the model processes cardiac waveforms. But once these interpretable features flow into the deeper, fully connected layers, it becomes harder to follow how they influence the final prediction. That is where our analysis comes in. We set out to trace how feature importance shifts across VPNet’s layers and to see whether the model’s overall decision-making process remains understandable from a clinical perspective. By pairing SHAP-based explanations with GPT-generated clinical narratives, we offer a clearer picture of how VPNet reasons, and how that compares to more conventional deep learning models.

Chapter 3

Methodology

3.1 Models

This study evaluates and compares three distinct neural network architectures (Table 3.1) for ECG classification. Each models represents a different strategy for extracting and processing features from time-series data.

Model	Input	Feature Extraction	Intermediate Representation	Output
VPNet	ECG	VP Layer (Hermite)	VP Coeffs	Softmax
CNN	ECG	Conv + Pool	Flattened Filters	Softmax
FCNN	ECG	–	Dense Layers	Softmax

Table 3.1: Model comparison of input, feature extraction method, intermediate representation, and output layer

3.1.1 Variable Projection Network (VPNet)

VPNet integrates the mathematical framework of variable projection into a neural network to enhance interpretability and domain alignment. Its architecture is structured of two primary components.

- VP Layer: This layer projects the input ECG signal onto a set of parameterized basis functions. It performs three key operations:

1. Projects the signal orthogonally onto a latent function space.
2. Extracts coefficients of the projection that represent the signal in the basis.

These extracted values, referred to as VP coefficients, quantify how much

each basis function (e.g., each Hermite wave) contributes to reconstructing the original ECG signal. Each coefficient captures a specific morphological component, such as peak height or asymmetry, thus forming a compact, interpretable representation of the heartbeat.

3. Learns the nonlinear parameters of the basis functions during training.

- Fully Connected Layers: Following the VP Layer, these layers map extracted features to class probabilities.

Mathematically, the output of the VP Layer is defined as:

$$f_{vp}(x) := \Phi^+(\theta)x = c \quad \text{with} \quad x \in \mathbb{R}^m$$

Here $\Phi^+(\theta)$ denotes the Moore-Penrose pseudoinverse of the function matrix $\Phi(\theta)$ includes the nonlinear parameters (e.g., translation and dilation of the basis functions), and c represents the learned feature coefficients (Figure 3.1).

For the function basis, we employ an adaptive Hermite system, which is particularly suitable for modeling ECG signals due to its efficiency in capturing wave-like morphologies. The basis functions are defined as:

$$\phi_k(t; \tau, \lambda) = \sqrt{\lambda} \Phi_k(\lambda(t - \tau))$$

where τ controls the temporal position and λ the scale (width) of the basis functions. These parameters offer direct physiological interpretability, with τ corresponding to wave timing and λ reflecting duration and shape.

We implemented two configurations of VPNet:

- VPNet with 8 coefficients ($n_{vp} = 8$)
- VPNet with 4 coefficients ($n_{vp} = 4$)

Both versions include a single hidden layer with 16 neurons, followed by a ReLU activation and a softmax output layer.

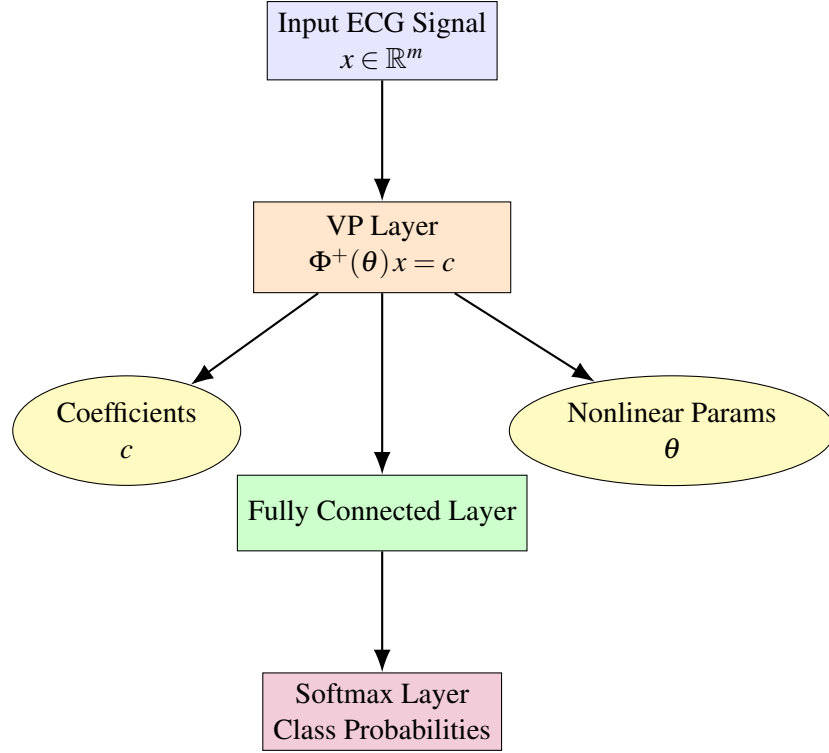


Figure 3.1: Architecture of VPNet showing the VP layer with Hermite projections, resulting coefficients and nonlinear parameters, followed by fully connected and output layers.

3.1.2 Convolutional Neural Network (CNN1D)

As a baseline deep learning model, we implemented a one-dimensional convolutional neural network with the following structure:

- Input layer for raw ECG signals
- Convolutional layer with 8 filters and a kernel size of 15
- ReLU activation
- Max pooling with pool size of 2
- Flattening layer
- Fully connected layer with 16 neurons and ReLU
- Softmax output layer

This model learns relevant temporal patterns directly from the signal, without relying on predefined basis functions.

3.1.3 Fully Connected Neural Network (FCNN)

The FCNN model serves as a simple baseline that operates directly on flattened ECG inputs. Its architecture consists of:

- Flattened input layer
- Two hidden layers, each with 16 neurons and ReLU activation
- Softmax output layer

Unlike VPNet and CNN, this model does not perform explicit feature extraction, relying instead on dense representations of the raw input.

3.1.4 Model Complexity and Parameter Counts

In addition to architectural differences, the total number of trainable parameters serves as a useful indicator of each model’s representational capacity and computational footprint. The following table (Table 3.2) summarizes the parameter counts for each implemented model.

Model	Number of Trainable Parameters
VPNet (4 coefficients)	180
VPNet (8 coefficients)	116
FCNN	5,682
CNN1D	1,922

Table 3.2: Total trainable parameters across model architectures

These values show the underlying structural choices of each architecture. VPNet’s parameter efficiency comes from its explicit use of Hermite basis projections in the VP layer, which serves as a compact feature extractor before passing its output to a shallow fully connected classifier. In contrast, the FCNN and CNN1D models rely on broader learned representations across deeper layers, resulting in higher parameter counts.

While parameter size alone does not determine performance or interpretability, it provides useful context when assessing the computational and memory requirements of each architecture throughout subsequent analysis.

3.2 Dataset

3.2.1 MIT-BIH Arrhythmia Database

All models were trained and evaluated using the MIT-BIH Arrhythmia Database, a widely used benchmark in ECG classification. The dataset contains 48 half-hour two-channel ECG recordings from 47 patients, sampled at 360 Hz with 11-bit resolution across a 10 mV range.

Following the setup described by Kovács et al. [29], we framed the task as a binary classification problem by distinguishing between normal heartbeats and ventricular ectopic beats (VEBs). However, as noted in previous studies, the original dataset is highly imbalanced, with a strong predominance of normal beats over pathological ones. This imbalance introduces a challenge for training models that generalize well across both classes.

To address this, we prepared a balanced variation of the dataset by undersampling the majority class (normal beats), following the recommendations of de Chazal et al. [31]. This resampling approach improves class balance during training and increases the robustness of the models against bias toward the dominant class.

We used the following predefined training and test sets:

- `ecg_train.mat`: Contains labeled heartbeats for training
- `ecg_test.mat`: Contains separate labeled heartbeats for evaluation

3.2.2 ECG Waveform Regions

Understanding the structure of the electrocardiogram (ECG) waveform is essential for interpreting both clinical diagnoses and model behavior in ECG classification tasks. The ECG signal reflects the electrical activity of the heart over time and consists of several morphologically distinct regions, each corresponding to a specific phase in the cardiac cycle.

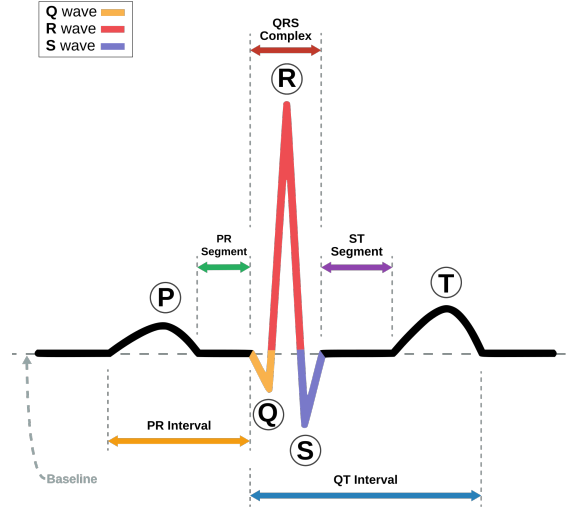


Figure 3.2: Annotated ECG waveform showing key regions (P wave, PR interval, QRS complex, and T wave)

As illustrated in Figure 3.2, the main components of a typical heartbeat are as follows:

- **P wave:** Represents atrial depolarization, typically occurring at the beginning of the heartbeat. It is relatively low in amplitude and precedes the QRS complex.
- **PR interval:** Spans from the beginning of the P wave to the beginning of the QRS complex, reflecting the time taken for electrical conduction from the atria to the ventricles.
- **QRS complex:** Corresponds to ventricular depolarization and is the most prominent and high-frequency component of the ECG. It typically occurs over a short time interval and is crucial for heartbeat detection algorithms.
- **T wave:** Reflects ventricular repolarization. It occurs after the QRS complex and is usually broader and lower in amplitude.

These regions are physiologically meaningful and are routinely analyzed by cardiologists to detect abnormalities, such as arrhythmias or ischemic events. In this study, they also serve as a crucial part for interpreting feature attributions via SHAP. Each time-indexed feature in our input window is aligned with one of these waveform segments based on its position relative to the R-peak. This mapping allows us to translate technical attributions into clinically interpretable insights.

Moreover, during SHAP-based analysis, we aggregate feature importances across these regions to understand which part of the ECG the model relies on most for its predictions. This wave-region mapping is further leveraged in the GPT explanation pipeline, where it provides the temporal context necessary for generating clinically grounded narratives.

3.2.3 Preprocessing and Segmentation

Each heartbeat was segmented using a fixed-length window of 100 samples (approximately 0.28 seconds), centered on the R-peak. This window size captures the full QRS complex and portions of both the P and T waves.

The preprocessing pipeline included:

- R-peak detection and alignment at the center of the window
- Amplitude normalization to reduce inter-patient variability
- Resampling to ensure uniform signal length across all samples

3.2.4 Sampling Rate Alignment

Although the MIT-BIH Arrhythmia Database is sampled at a nominal rate of 360 Hz, our preprocessing pipeline segments each ECG beat into a fixed-length window of 100 samples, spanning approximately 0.28 seconds. This results in an effective sampling rate of roughly 357.1 Hz.

While this approximation introduces a minor deviation from the official rate, it remains within an acceptable margin of error and does not impact the model’s performance or SHAP-based analysis. More importantly, defining an effective sampling rate enables accurate mapping of input indices to time in milliseconds. This mapping classifies each input feature into physiologically meaningful wave regions, such as the P-wave, QRS complex, or T-wave/ST segment, based on its relative position within the beat. This alignment serves several critical purposes:

- It enhances the interpretability of SHAP values by assigning each feature a clinically relevant temporal label.
- It improves the quality of GPT-generated natural language explanations, which rely on precise temporal references when describing model behavior.

- It ensures consistency across models when comparing feature importance at the raw input level, despite architectural differences in feature extraction.

By aligning feature indices with the physiological structure of the heartbeat, we bridge the gap between statistical model output and clinically meaningful interpretation, which is crucial for trustworthy ECG classification.

3.3 SHAP Computation

3.3.1 GradientExplainer

To quantify feature contributions across model architectures, we employed SHAP’s GradientExplainer. This method is specifically tailored for differentiable models such as neural networks. It computes SHAP values, i.e., the marginal contribution of each input feature, using a path-integrated version of the input gradients.

Formally, GradientExplainer implements the Integrated Gradients method [32], which approximates feature attributions by integrating the gradients of the model output with respect to its inputs along a linear interpolation path between a baseline input $x^{(0)}$ and the actual input x . For a feature i , the SHAP value ϕ_i is approximated as:

$$\phi_i(x) = (x_i - x_i^{(0)}) \cdot \int_{\alpha=0}^1 \frac{\partial f(x^{(0)} + \alpha \cdot (x - x^{(0)}))}{\partial x_i} d\alpha$$

where f denotes the model’s output function. This formulation satisfies key axioms of attribution, such as local accuracy and implementation invariance, and ensures consistency across deep learning layers.

In our implementation, the baseline input $x^{(0)}$ is taken from a representative subset of the test set (see Section 3.2). The actual SHAP explainer is initialized (Code 3.1) as shown in the following:

```
explainer_vp_model_8 = shap.GradientExplainer(vp_model_8_shap,
↪ background_data)
```

Code 3.1: SHAP explainer initialization

Each model wrapper was paired with its own SHAP explainer. For example, the following table (Table 3.3) shows how SHAP is applied to several layers of VPNet.

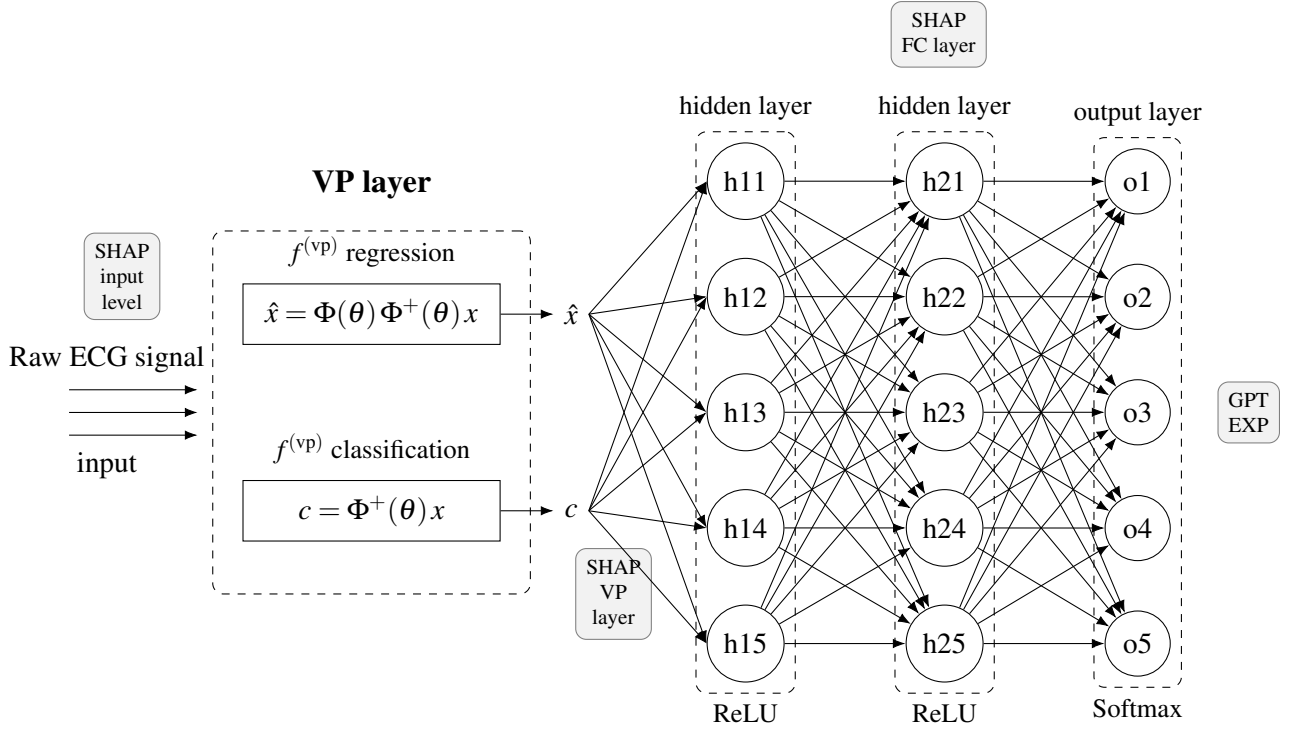


Figure 3.3: Integrating SHAP on VPNet architecture.

3.3.2 Background and Test Data

The background data used to define a reference distribution consisted of the first 100 samples from the test set (Code 3.2). SHAP values were computed on a subset of 10 test samples:

```
1 background_data = x_test[:100]
2 test_data = x_test[:10]
```

Code 3.2: Background and test data setup

This setup provided a computationally efficient way to generate attributions while ensuring a representative baseline.

3.3.3 Interpreting SHAP Values

The resulting SHAP values quantify how much each input feature pushes the model's output away from the expected value at the baseline. Positive values indicate that a feature increases the predicted probability of the target class, while negative values indicate a suppressive effect.

Importantly, SHAP values are additive which means the model prediction $f(x)$ can be decomposed into the sum of all feature attributions and the baseline prediction $f(x^{(0)})$:

$$f(x) = f(x^{(0)}) + \sum_{i=1}^n \phi_i(x)$$

This property ensures that the entire prediction can be explained as the sum of individual feature contributions which is enhancing model transparency.

3.3.4 Aggregating SHAP Values

For each model and layer, we computed SHAP values and summarized feature importance (Code 3.3) by averaging the absolute SHAP values across all test samples:

```

1 mean_abs_shap = {
2     'VPNet (8 coeffs)':
3     ↪ np.abs(extract_shap_arr(vp_shap_values)).mean(axis=0),
4     # other models...
5 }
```

Code 3.3: Computing SHAP values and summarizing feature importance

This aggregation produces a robust metric that reflects both positive and negative contributions to the model output.

3.4 GPT-Based Explanation Generation

To enhance the interpretability of SHAP results, we implemented GPT-4o-mini to generate textual explanations grounded in clinical reasoning.

3.4.1 Structured Function Interface

We defined a structured schema (Code 3.4) to standardize the explanation format, requiring each output to include the model name, layer, target class, top, features, and accompanying clinical rationale:

```

1 {
2     "name": "explain_shap_features",
```

```

3  "description": "Explain SHAP-based model decision for ECG
   ↪  classification.",
4  "parameters": {
5      "type": "object",
6      "properties": {
7          "model_name": {"type": "string"},
8          "layer_name": {"type": "string"},
9          "class_index": {"type": "integer"},
10         "insights": {
11             "type": "array",
12             "items": {
13                 "type": "object",
14                 "properties": {
15                     "feature": {"type": "string", "enum":
16                         ↪  [f["feature"] for f in top_features]},
17                     "importance": {"type": "number"},
18                     "direction": {"type": "string", "enum":
19                         ↪  ["increases", "decreases"]},
20                     "rank": {"type": "integer"},
21                     "clinical_reasoning": {"type": "string"}
22                 },
23                 "required": ["feature", "importance",
24                     ↪  "direction", "rank", "clinical_reasoning"]
25             }
26         },
27         "required": ["model_name", "layer_name", "class_index",
   ↪  "insights"]
28     }
29 }

```

Code 3.4: Structured function schema for GPT-based explanation output

3.4.2 Prompt Design and Safeguards

To guide the language model, we provided a domain-specific system prompt (Code 3.5) emphasizing evidence-based interpretation and appropriate uncertainty:

```
1 system_prompt = (  
2     "You are an expert in biomedical signal analysis and explainable  
   ↳ AI."  
3 )  
4  
5 user_prompt = f"""  
6 You must only use the provided model_name, layer_name, class_index,  
   ↳ and insights-do not alter them or invent new ones.  
7  
8 Explain how the model {model_name} makes a decision for class  
   ↳ {class_index} at the {layer_name} level,  
9 based only on the following features, in the exact order and names  
   ↳ given:  
10  
11 {features_json}  
12  
13 Do not invent or rename any features.  
14  
15 The features are listed in order of descending importance (rank 1 is  
   ↳ the most important).  
16 Each insight object includes 'direction' (whether the feature  
   ↳ increases or decreases the logit)  
17 and 'rank'.  
18  
19 Also, explicitly compare how the feature with rank 2 differs from  
   ↳ the feature with rank 1  
20 in terms of their effect on the model's decision.  
21  
22 Use AHA terminology and evidence-based clinical reasoning.
```

```
23 Do not invent conditions or recommendations without evidence.  
24 If evidence is unclear, write 'No clear data to determine...'.  
25 """
```

Code 3.5: Defining a domain-specific system prompt

The prompt explicitly discouraged speculation, instructed the model to avoid referencing full clinical intervals unless appropriate, and required clear acknowledgment when evidence was insufficient.

3.4.3 Feature Selection

We selected the top-K features (Code 3.6) based on their average absolute SHAP values:

```
1 mean_abs = np.abs(shap_values).mean(axis=0)  
2 top_idx = mean_abs.argsort()[-top_k:] [::-1]
```

Code 3.6: Selecting top-K features of SHAP values

This ensured that the generated explanations focused on the most influential input regions.

3.4.4 Generation Pipeline

For each model and layer, the selected features and importance scores were passed to the GPT model (Code 3.7) via function calling:

```
1 response = client.chat.completions.create(  
2     model="gpt-4o-mini",  
3     messages=messages,  
4     functions=[function_schema],  
5     function_call={"name": "explain_shap_features", "arguments":  
6         ↪ json.dumps(func_args) },  
7     temperature=0.1,  
8     max_tokens=2048,  
9 )
```


Code 3.7: Generating the response from GPT

We used a low temperature setting to ensure consistency and factual stability across generated explanations.

By combining SHAP with domain-grounded text generation, this pipeline produced layered, clinically relevant narratives that explain not just what features were important, but why they matter from a medical perspective. These explanations enable side-by-side comparison of architectural behavior and interpretability in a way that aligns with clinical reasoning.

Chapter 4

Results

4.1 SHAP Feature Attribution Analysis

As detailed in Section 3.2.4, each ECG beat was segmented into a fixed-length window of 100 samples to enable consistent model input and meaningful temporal alignment. This alignment with physiological wave regions (e.g., P-wave, QRS complex, T-wave) provides the foundation for clinically interpretable SHAP analysis and GPT-based explanation.

The resulting SHAP outputs revealed distinct patterns of feature attribution across model architectures, with particularly informative insights observed in the individual VP coefficients.

4.1.1 Model-Level Feature Attribution Patterns

CNN1D and FCNN Models

The analysis results (Figure 4.1) in these both baseline architectures showed highly variable SHAP values across ECG regions with both positive and negative contributions (Section 3.3.3). Feature importance was diffusely distributed with SHAP values ranging between -0.02 and 0.02.

Both the CNN1D and FCNN models fairly showed scattered attribution patterns across different parts of the ECG signal, without any particular region standing out as clearly driving the predictions. This suggests that their internal logic may not be strongly aligned with clinical reasoning.

For the CNN1D model, we noticed that several QRS complex features, such as those around 112.0 ms, 117.6 ms, and 126.0 ms, tended to have clusters of high SHAP values. These clusters hint at some localized influence of QRS duration on the model's output. However, this influence was not consistent in direction and was often mixed with contributions from other regions, such as the T-wave, making the attribution pattern harder to interpret.

On the other hand, the FCNN model displayed a more even pattern, with positive attributions concentrated mostly in the T-wave and ST segment areas, especially around 142.8 ms, 165.2 ms, and 145.6 ms. This might indicate that the model leans more heavily on repolarization features when making decisions, although these attributions do not necessarily reflect strong clinical discriminability.

Taken together, these results point to a shared limitation in both models which they do not seem to rely on a focused or clinically interpretable set of features. Compared to more structured models in the upcoming discussion, for example VPNet, both CNN1D and FCNN may lack the capacity to learn physiologically meaningful representations from the data.

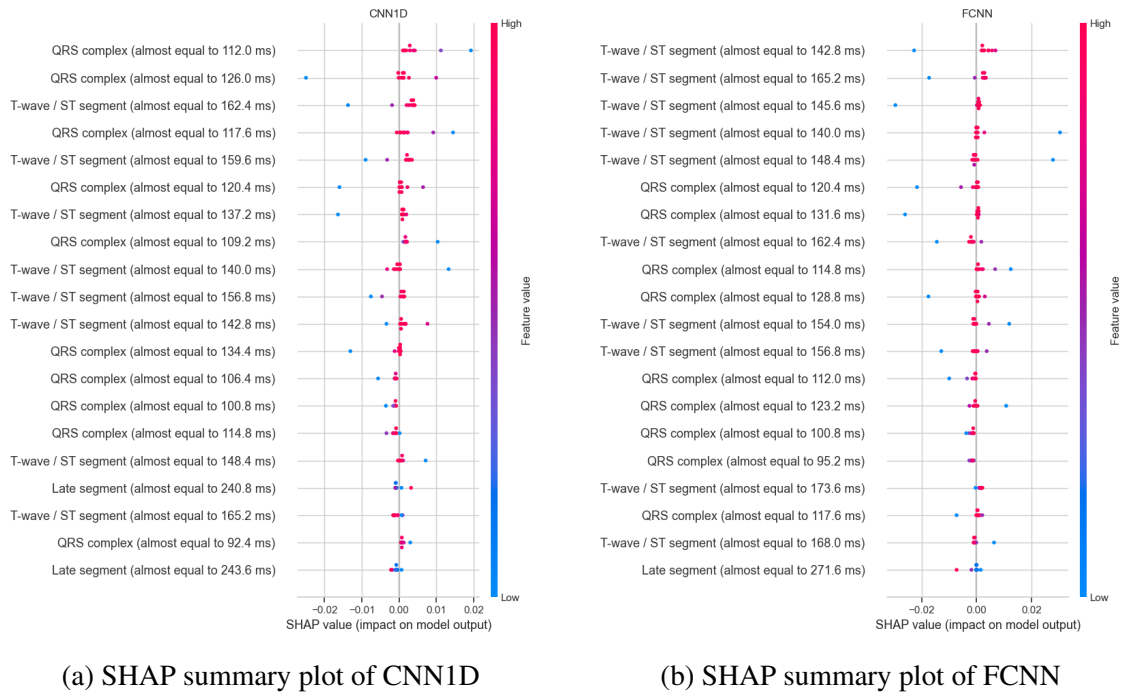


Figure 4.1: Comparison of SHAP summary plot between CNND1D and FCNN models

Raw ECG Baseline

Compared to the neural network models, the raw ECG input produced a distinctly different SHAP attribution pattern. Most of the features showed strongly negative SHAP values (Figure 4.2), ranging from approximately -4 to 0, indicating that higher values in these features tend to reduce the model’s predicted probability for the class of interest.

The magnitude of these SHAP values was noticeably larger than in any other model, suggesting a more decisive influence of individual input points on the model’s output. Moreover, the tight clustering of SHAP values per feature reflects a high level of consistency across samples, pointing to a stable attribution pattern.

Interestingly, there appears to be a temporal gradient where features corresponding to later points in the waveform (i.e., higher ms values, particularly in the T-wave and ST segment region) showed increasingly negative contributions. This could imply that the model is particularly sensitive to repolarization dynamics that is represented by the T-wave when using raw signal data, although the physiological implications of such a pattern would require further clinical interpretation.

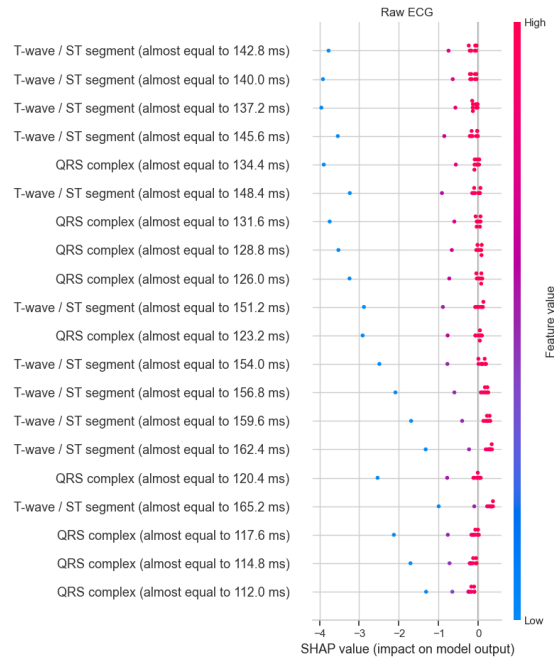


Figure 4.2: SHAP summary plot of raw ECG baseline

VPNet Models

The VPNet models showed distinct attribution behaviors compared to the CNN, FCNN, and raw ECG baselines, highlighting the effect of compressed Hermite representa-

tions on feature salience. Although VPNet models operate on compressed representations (4 or 8 Hermite coefficients), we performed SHAP attribution by mapping these internal representations back to the original input space. This inverse mapping allows us to visualize feature importance over the full 100-sample input window, which is why the SHAP summary plots display attributions for all input features rather than just the coefficients.

- VPNet (4 coefficients): As shown in the following figure (Figure 4.3), this model showed predominantly negative SHAP values, ranging from approximately -0.15 to 0, with T-wave and ST segment features dominating the most influential contributors. Several QRS complex features, notably around 110–115 ms, stood out for having both positive and negative attributions, indicating a more divergent influence across different samples. While the overall SHAP values were smaller than those from the raw ECG model, they were generally larger in magnitude than those observed in CNN1D and FCNN, suggesting a stronger and more decisive attribution pattern.
- VPNet (8 coefficients): In contrast, the 8-coefficient variant (Figure 4.3) produced more moderate SHAP values, mostly between -0.04 and 0.02. The attributions appeared more evenly distributed between positive and negative directions, reflecting a balanced impact of features. T-wave and ST segment features in the later part of the signal (e.g., 159–168 ms) showed notably stronger negative contributions. This version also showed a finer-grained differentiation between time points within the same physiological region, particularly in the repolarization phase, indicating a more nuanced internal representation.

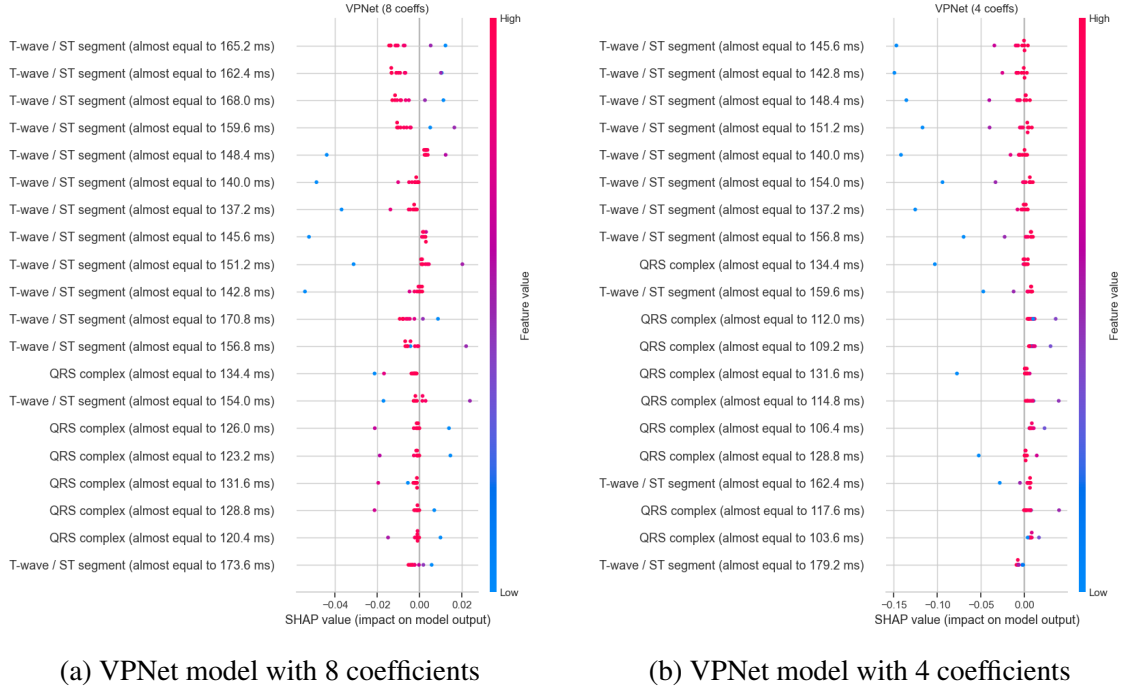


Figure 4.3: Comparison of SHAP summary plot between VPNet model with 8 and 4 coefficients

Together, these findings suggest that the 4-coefficient VPNet produces sharper and more polarized attributions, while the 8-coefficient version favors more stable and distributed explanations, potentially reflecting a trade-off between compression and interpretive smoothness.

An interesting observation across models was attribution consistency, measured by the tightness of point clusters in the SHAP plots. The Raw ECG and VPNet models exhibited high consistency, as seen in the compact vertical spread of SHAP values per feature, indicating stable and repeatable feature importance across samples. In contrast, the CNN1D and FCNN models showed more scattered distributions, suggesting less reliable feature utilization and potential instability in how input features contribute to predictions.

4.1.2 Individual VP Coefficient Analysis

One of the key contributions of this study is our deep-dive into how individual VP coefficients (Section 3.1.1) contribute to the model’s decisions. By applying SHAP at the coefficient level, we were able to uncover how specific compressed representations relate to distinct ECG waveform regions (Figures 4.4), offering a rare glimpse into the inner workings of VPNet’s learned features.

4. Results

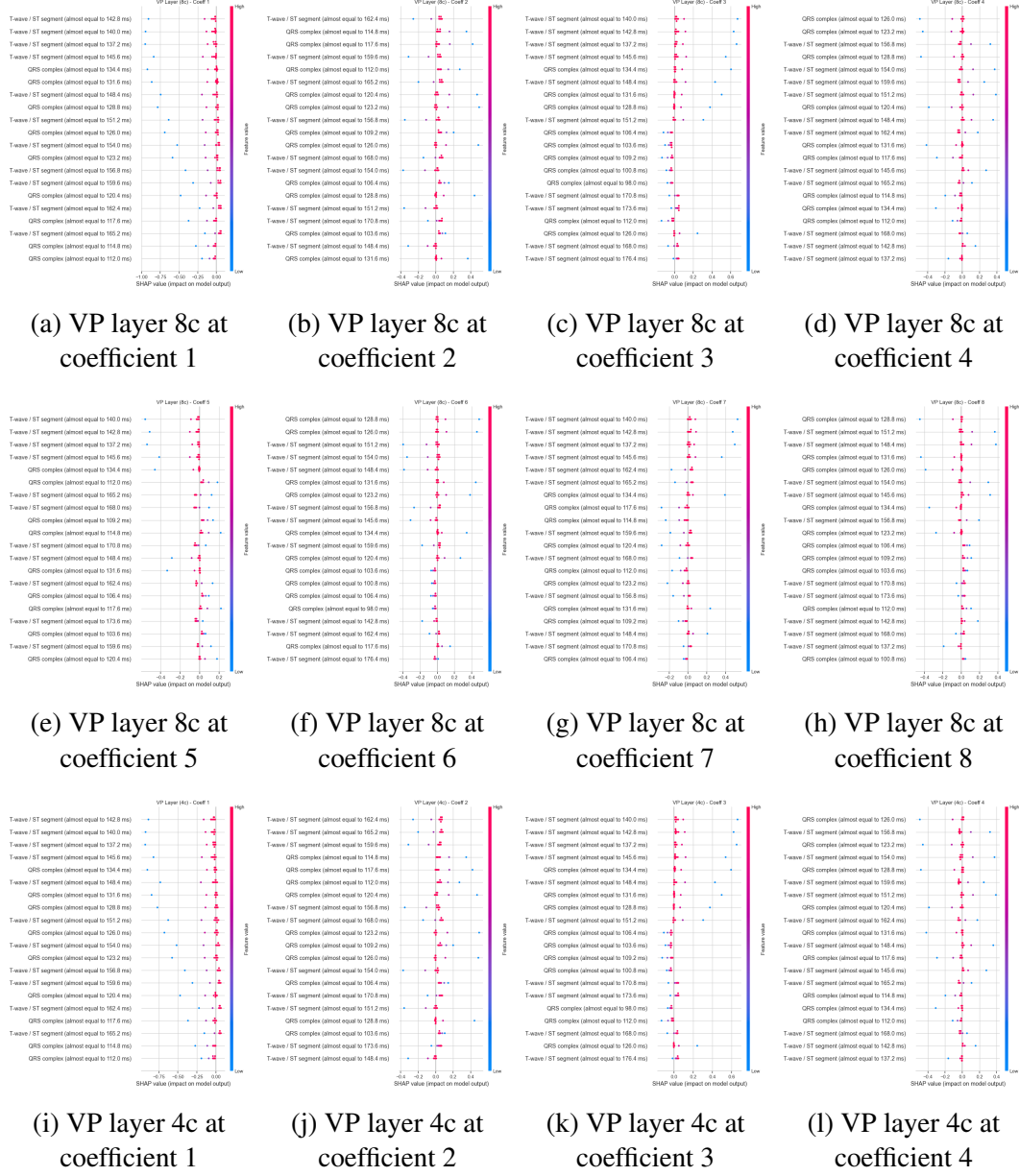


Figure 4.4: SHAP summary plots for VP layer coefficients (8c and 4c)

Coefficient Specialization

Across both the 4-coefficient (4c) and 8-coefficient (8c) VPNet configurations, a clear pattern of specialization emerged on each coefficient, showing a tendency to focus on particular ECG phases or signal characteristics.

- **Coefficient 1:** In both 4c and 8c models, this coefficient stood out for its strong and consistent negative contributions across nearly all ECG features. SHAP values ranged from around -0.75 to 0 in the 4c model and -1.0 to 0 in the 8c model. The most pronounced patterns were seen in the T-wave/ST segment region (159–165 ms), which repeatedly appeared among the top features. Interestingly, the pattern remained consistent across samples, suggesting that Coefficient 1 reliably encodes the overall morphology of the ECG waveform. The near-identical behavior between the two model versions implies that this fundamental representation is learned robustly, regardless of model complexity.
- **Coefficient 2:** Unlike Coefficient 1’s uniform negativity, Coefficient 2 showed a balanced mix of positive and negative attributions (-0.4 to 0.4), reflecting a more nuanced role. Here, QRS complex features, especially around 110–120 ms, exhibited distinctively positive SHAP values, while later T-wave features tended to have negative contributions. This contrast suggests that Coefficient 2 may specialize in capturing phase transitions and discriminating between morphological regions, rather than encoding the full waveform.
- **Coefficient 3:** Coefficient 3 showed a nearly inverted behavior compared to Coefficient 1, with predominantly positive SHAP values (0 to 0.6). It consistently showed negative contributions for early QRS features (103–109 ms), but strong positive contributions for later T-wave/ST segment features. This split implies that Coefficient 3 plays a more targeted role in capturing repolarization characteristics and later-stage waveform details.
- **Coefficient 4:** The fourth coefficient offered a more ambiguous contribution. SHAP values were tightly clustered around zero, with both models showing weaker and less consistent attributions. While the 8c model hinted at some relevance in the 123–126 ms QRS region, the 4c variant contributed little overall. These patterns suggest Coefficient 4 functions as a residual encoder, picking up information not strongly represented by the earlier coefficients.

- **Higher-Order Coefficients (8c only):** In the 8-coefficient model, Coefficients 5 through 8 showed a progressive drop-off in importance. Most of their SHAP values hovered near zero, with only occasional spikes of activity (e.g., Coefficient 5’s contribution to the 112.0 ms QRS feature). These coefficients appear to encode minor signal variations, offering minimal additional physiological insight. The sparsity observed here supports the idea that most diagnostically relevant information is already captured in the first few coefficients.

Progressive Information Encoding

When comparing across coefficients, a clear hierarchical structure became evident:

1. Coefficients 1–3 carried the majority of the predictive power, encoding the ECG’s core morphology and clinically relevant patterns.
2. Coefficient 4 contributed marginally, likely handling leftover or non-dominant signal components.
3. Coefficients 5–8 (in the 8c model) held diminishing importance, encoding residual noise or fine-grained details with little impact on prediction.

This layered encoding aligns well with clinical reasoning where physicians first evaluate the overall ECG shape, then zoom into specific waveform regions, and finally look for subtler anomalies when necessary.

Comparison Between 4c and 8c Models

Interestingly, the first three coefficients in both models showed strikingly consistent behavior, suggesting that the core ECG structure is learned reliably regardless of the number of coefficients. However, the 4c model, having fewer coefficients to work with, appeared to pack more information into each one, leading to larger SHAP magnitudes per coefficient. In contrast, the 8c model spread the representation more thinly, with its higher-order coefficients handling residual nuances.

This analysis helps explain why the 4c model performs nearly as well as the 8c model, despite having half the number of coefficients. Most of the meaningful signal is already captured by the first three or four, and beyond that, additional coefficients provide diminishing returns in terms of clinical insight or model performance.

4.1.3 Clinical Relevance of Coefficient Specialization

One of the most compelling aspects of the VPNet model is how naturally its internal representations seem to align with established clinical practices in ECG interpretation.

1. Coefficient 1, which consistently encodes the overall shape of the ECG waveform, mirrors the first glance that clinicians take when scanning for rhythm irregularities or gross abnormalities.
2. Coefficient 2 focuses heavily on the QRS complex, particularly between 110 and 120 ms, a region of high diagnostic importance when evaluating ventricular depolarization, often used to detect conduction delays or arrhythmias.
3. Coefficient 3 centers on the T-wave and ST segment, corresponding with clinical assessment of repolarization, which is key in identifying signs of ischemia, electrolyte imbalances, or myocardial injury.

This kind of structural correspondence between how VPNet organizes features and how cardiologists interpret ECGs offers a promising bridge for making deep learning models more explainable in medical settings. It suggests that VPNet is not just learning statistical patterns, but it may be capturing physiologically meaningful representations that resonate with real-world clinical reasoning.

Moreover, the progressive drop in importance from Coefficient 1 to the higher-order coefficients mirrors how clinicians prioritize features, starting with broad patterns before diving into finer-grained waveform components. The consistent importance of T-wave/ST segment features across multiple coefficients reinforces their known diagnostic value and reflects the model’s ability to internally emphasize relevant clinical information.

4.1.4 Physiological Significance of Attribution Patterns

When comparing across models, one consistent trend stood out where T-wave and ST segment features, especially those in the 137 to 170 ms range, were among the most influential. This aligns well with how these regions are prioritized in ECG interpretation due to their diagnostic relevance.

However, different models treated ECG components very differently:

1. VPNet models displayed more clinically interpretable attribution patterns, clearly separating QRS complex and T-wave regions, and preserving their distinct physiological roles.
2. CNN1D and FCNN models, on the other hand, had more diffuse and scattered attributions, with less obvious alignment to known diagnostic regions, making their internal logic harder to connect with clinical reasoning.
3. The raw ECG baseline offered a different perspective, it showed strong negative SHAP values across both QRS and T-wave regions, with a temporal progression that might reflect the natural information distribution across the ECG cycle.

Overall, VPNet’s structure appears to better retain clinically salient features in a way that resonates with how human experts read ECGs. This supports the broader claim that its compressed representation approach yields not just compact features, but also ones that are more aligned with medical understanding.

4.1.5 Impact of Coefficient Number in VPNet

The comparison between the 4-coefficient and 8-coefficient VPNet variants also yielded valuable insights into model behavior:

- The 4-coefficient model showed generally larger SHAP magnitudes, indicating that fewer coefficients were responsible for carrying more concentrated information. This suggests a more compact and interpretable representation.
- The 8-coefficient model, while having lower SHAP magnitudes overall, revealed a more nuanced distribution of importance, especially among T-wave/ST segment features at various time points. This model appears to trade compactness for greater resolution in how it encodes ECG features.
- Both versions showed high attribution consistency, reflected in the tight clustering of SHAP values, but their contribution patterns, in terms of positivity or negativity, were distributed differently.

These findings point to a meaningful trade-off where the 4c model offers interpretability and simplicity, which is valuable in clinical settings, while the 8c model provides more detailed and granular representation, which could be beneficial for detecting subtle

pathologies. Choosing between the two may ultimately depend on the desired balance between clinical transparency and diagnostic depth.

4.2 Model-Wide SHAP Feature Importance Patterns

To complement the insights provided by SHAP summary plots, we also conducted a model-wide analysis of SHAP feature importance by averaging the absolute SHAP values across all test samples for each model. This approach offers a global perspective on how different models and layers distribute importance across the ECG time series.

4.2.1 Mean SHAP Value Distributions Across Models

Figure 4.5 presents the mean absolute SHAP value distributions for each model, aligned by timepoint (i.e., feature index in the ECG signal). This allows us to see direct comparison of where each architecture focuses its attention and how attribution patterns vary with model design.

VPNet vs. CNN/FCNN Models

The VPNet variants, both with 4 and 8 coefficients, demonstrated clear, focused importance peaks centered around the QRS complex region (roughly feature indices 40–60), with particularly pronounced activity around index 50. This region aligns with the ventricular depolarization phase and is a clinically significant marker in ECG analysis. The 4-coefficient model exhibited slightly sharper peaks and higher SHAP magnitudes than the 8-coefficient version, suggesting that with fewer coefficients, the model concentrates importance more intensely on key features.

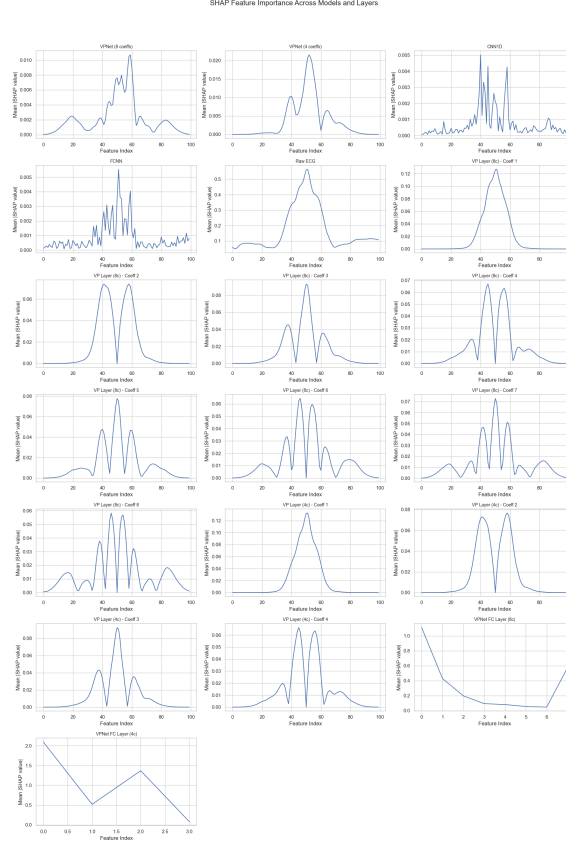


Figure 4.5: Mean absolute SHAP values across ECG timepoints for each model.

In contrast, both the CNN1D and FCNN models showed highly diffuse attribution patterns. Their SHAP distributions were scattered across a wide range of features with multiple small peaks rather than a dominant region of focus. This suggests that these models rely on many small contributions from across the signal, rather than prioritizing a focused, physiologically meaningful region. These findings are consistent with earlier SHAP summary plot observations and highlight a lack of diagnostic clarity in the CNN/FCNN attention strategies.

Raw ECG Baseline

The raw ECG baseline model produced a broad, bell-shaped SHAP distribution centered around index 50, again corresponding to the QRS complex. Unlike the learned models, the raw ECG attribution was smooth and symmetrical, with substantially larger SHAP magnitudes. This likely reflects the natural statistical prominence of the QRS complex in the uncompressed signal, prior to any model transformations. Interestingly, the baseline model’s shape closely resembled Coefficient 1 of VPNet, suggesting that VPNet retains and builds upon this core waveform representation.

4.2.2 VP Coefficient-Level Patterns

Examining SHAP values at the VP layer level revealed distinct attribution patterns across individual coefficients, especially in the 8-coefficient VPNet model. Each coefficient appeared to specialize in representing different waveform structures:

- **Coefficient 1** showed a smooth, centered peak over the QRS region, mirroring the raw ECG baseline and encoding the fundamental morphology of the signal.
- **Coefficient 2** displayed a bimodal distribution with two peaks flanking the QRS complex (indices 40 and 60) and a valley near the center. This pattern suggests sensitivity to transitions before and after the depolarization phase.
- **Higher-order coefficients (3–8)** exhibited more complex, irregular patterns, with multiple peaks across mid-to-late signal regions, possibly capturing finer aspects of repolarization or noise.

The 4-coefficient model showed similar patterns but with higher SHAP magnitudes per coefficient, indicating more concentrated encoding due to limited representation capacity.

4.2.3 Fully Connected Layer Attribution to Coefficients

The SHAP values of the fully connected layer weights revealed how the final decision depends on each VP coefficient. As shown in Figure 4.5, importance was strongly front-loaded, with the first few coefficients contributing the most to the model’s output:

- In both 4c and 8c models, Coefficient 1 received the highest attribution.
- The 4c model showed a sharp importance peak at Coefficient 2, while the 8c model had a more gradual decay.
- Coefficients 5–8 in the 8c model contributed minimally, with near-zero SHAP values.

This trend supports the hypothesis that VPNet forms a sparse and interpretable latent space, with the first few coefficients encoding the most diagnostically relevant information.

4.2.4 Cross-Model Comparison Summary

Our comparative SHAP analysis revealed important differences in how each model interprets and utilizes ECG features, both in terms of what they consider important and how consistently they do so.

1. **Attribution Patterns:** The VPNet models showed clear, physiologically coherent attribution, with distinct separation between the QRS complex and T-wave/ST segment. In contrast, the CNN1D and FCNN models produced more scattered and diffuse patterns, making their reasoning harder to interpret clinically.
2. **Attribution Magnitude:** The Raw ECG model produced the largest SHAP values, reflecting its direct access to uncompressed signal features. VPNet models came next in magnitude, followed by CNN1D and FCNN, which had notably lower attribution values. This hierarchy mirrors how much each model interacts with raw signal data.
3. **Attribution Consistency:** Both VPNet models and the Raw ECG baseline showed high attribution consistency, visible as tight clusters of SHAP values for each feature. CNN1D and FCNN, by contrast, showed more variability, suggesting less stable reliance on specific features.
4. **Attribution Consistency:** Both VPNet models and the Raw ECG baseline showed high attribution consistency, visible as tight clusters of SHAP values for each feature. CNN1D and FCNN, by contrast, showed more variability, suggesting less stable reliance on specific features.
5. **Physiological Alignment:** VPNet’s attributions aligned more closely with known diagnostic regions, demonstrating cleaner separation between depolarization (QRS) and repolarization (T-wave/ST). This structural alignment supports its clinical interpretability.

Taken together, these findings suggest that VPNet does not just match other models in classification performance, it does so by using a more compact, structured, and clinically meaningful feature space. The coefficient-level attribution further reveals how VPNet organizes this information in ways that mirror the logic of human ECG interpretation.

4.3 Correlation Analysis of Attribution Patterns

To quantify how similarly different models prioritize features, we computed pairwise Pearson correlations between the mean absolute SHAP values at the raw input level (Figure 4.6). This revealed a spectrum of attribution strategies:

- CNN1D showed the least agreement with other models (correlation $r \approx 0.49 - 0.63$), suggesting a more idiosyncratic way of interpreting input features.
- Raw ECG correlated strongly with both VPNet variants, $r = 0.87$ for the 4-coefficient model and $r = 0.80$ for the 8-coefficient model. This indicates that VPNet (4c) in particular retains much of the dominant signal structure from the raw input.

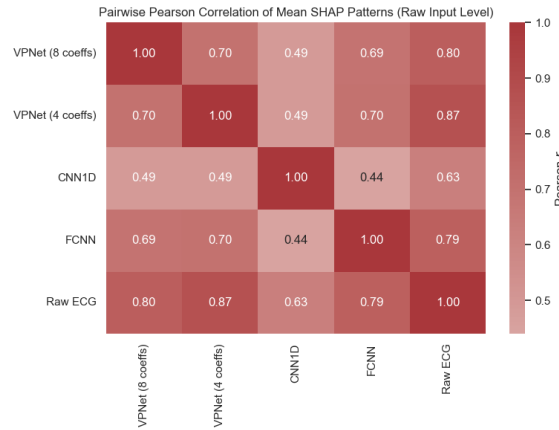


Figure 4.6: Pairwise pearson correlation of mean SHAP patternson raw input level

Interestingly, the correlation between the two VPNet versions was moderate ($r = 0.70$), showing that while they capture similar patterns, the number of coefficients influences how that information is distributed.

These trends were further illustrated in pairwise SHAP scatter plots (Figure 4.7), which highlighted large differences in attribution magnitude, with the Raw ECG model reaching SHAP values up to 0.5, while neural models clustered closer to 0.02. These non-linear relationships suggest that while neural models are transforming the input space, VPNet models still maintain some structured correspondence with raw signal features, unlike CNN1D and FCNN which appeared more inconsistent.

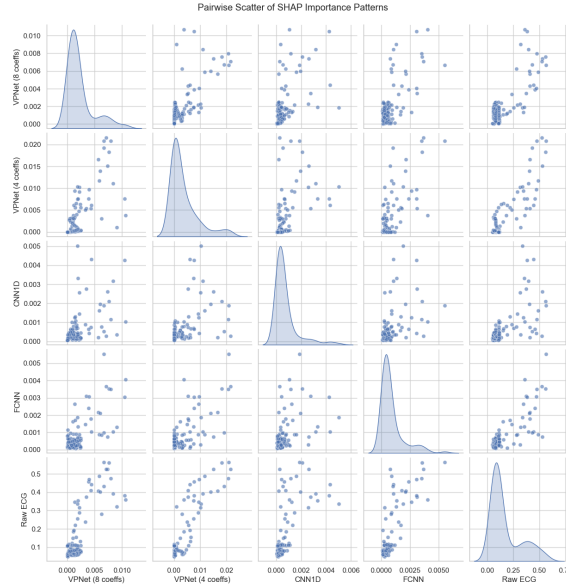


Figure 4.7: Pairwise scatter of SHAP importance patterns

4.3.1 Relationship Between VP Coefficients and Raw Signal

We also explored how each VP coefficient correlates with the raw signal’s SHAP values (Figure 4.8). The results were striking where all coefficients in both VPNet models showed strong positive correlations ($r > 0.7$) with the Raw ECG attribution pattern.

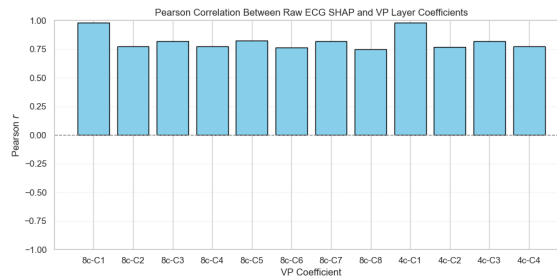


Figure 4.8: Pearson correlation between raw ECG SHAP and VP layer coefficients

- In particular, Coefficient 1 from both models (4c-C1 and 8c-C1) showed very strong alignment ($r \approx 0.98$) with the raw signal, suggesting that it captures the essential shape of the ECG waveform.
- Other coefficients (2–4) still showed high correlation but were slightly lower, indicating that they contribute more transformed, specialized representations of the signal.

This dual behavior, preserving raw signal structure while adding meaningful abstraction, is central to VPNet’s interpretability. Each coefficient encodes physiologically rel-

evant aspects of the ECG, but from a slightly different perspective, enriching the overall representation.

Notably, none of the coefficients showed negative correlation, reinforcing the idea that VPNet builds aligned, not inverted, representations of the raw signal. Coefficients with lower correlation still appear to reflect legitimate physiological components, just at a more abstract level.

4.3.2 Coefficient Importance in FC Layer

Analysis of SHAP importance in the fully connected layer (Figure 4.9) quantitatively confirmed our earlier observations about hierarchical information encoding in VP coefficients. Both models showed decreasing importance from Coefficient 0 to Coefficient 3, with the 4-coefficient model demonstrating substantially higher magnitude importance for Coefficient 0 ($r \approx 2.0$) compared to the 8-coefficient model ($r \approx 1.1$).

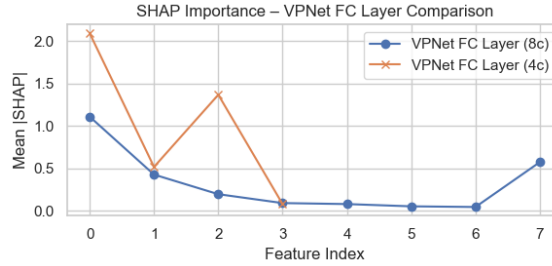


Figure 4.9: Comparison of SHAP importance at VPNet FC layer

The 4-coefficient model exhibited a distinctive peak at Coefficient 2 ($r \approx 1.4$), while the 8-coefficient model showed a more gradual decrease across coefficients 0-6, followed by an unexpected increase for Coefficient 7 ($r \approx 0.6$). Higher-order coefficients (4-6) in the 8-coefficient model showed near-zero importance, confirming their minimal contribution to the model’s decisions.

This pattern of coefficient importance provides quantitative evidence that VPNet models naturally prioritize information in the first 2-3 coefficients, with the 4-coefficient variant concentrating information more intensely per coefficient. This efficient encoding explains why the 4-coefficient model achieves comparable performance to the 8-coefficient model despite using half the representation dimensions.

The correlation and FC layer analyses together explain why VPNet (4c) achieves similar performance to the 8c variant with half the parameters, it concentrates the same information in fewer dimensions with stronger emphasis on raw signal characteristics. This

finding has important implications for model design in clinical applications, suggesting that smaller models may not only be more computationally efficient but potentially more interpretable while maintaining performance.

4.4 Post-Hoc Clinical Interpretation Using GPT

To enhance the clinical interpretability of SHAP attribution patterns, we employed GPT-4o-mini to translate SHAP outputs into structured, human-readable clinical narratives. For each model and VP coefficient, we provided GPT-4o-mini with the top five SHAP features (ranked by importance) and requested clinical interpretation in the context of ECG classification.

4.4.1 Model-Level Clinical Interpretation

GPT-generated clinical explanations revealed distinct interpretative patterns across model architectures. VPNet models consistently prioritized T-wave/ST segment features, particularly in the normal range (137-145 ms), with clear clinical reasoning that aligned these durations with normal ventricular repolarization. For example, for VPNet (8 coeffs), GPT reasoned that *"A T-wave/ST segment duration of approximately 140.0 ms is within the normal range, suggesting normal ventricular repolarization,"* while longer durations (>160 ms) were interpreted as *"slightly prolonged, which may indicate early signs of repolarization abnormalities."*

In contrast, CNN1D's explanations focused predominantly on QRS complex features, with clinical reasoning centered on conduction delays. For instance, its top feature (QRS complex at 112.0 ms) was interpreted as *"slightly prolonged, which can be indicative of a conduction delay,"* highlighting a fundamentally different focus than VPNet models. FCNN showed mixed focus between T-wave/ST segment and QRS complex features, resulting in less specialized and potentially more confusing clinical narratives.

An interesting finding was the strong alignment between attribution direction and clinical expectations across all models. Features within normal physiological ranges consistently increased the likelihood of normal classification, while abnormal durations decreased it. This biological plausibility in model decisions suggests VPNet is learning clinically relevant patterns rather than statistical artifacts.

4.4.2 VP Coefficient Clinical Specialization

GPT-generated explanations (Table 4.1) for individual VP coefficients provided striking evidence of physiological specialization. Clinical reasoning for Coefficient 1 in both 4c and 8c models consistently focused on normal T-wave/ST segments, interpreting these as indicators of *"normal ventricular repolarization"* and *"healthy cardiac function."*

Coefficient 2 in both models showed specialized focus on QRS complex features (particularly around 112-117 ms), with GPT reasoning that these durations indicate **"border-line conduction delay"** or *"incomplete bundle branch block."* This specialization aligns with clinical ECG interpretation, where QRS duration is a primary metric for assessing ventricular depolarization.

Coefficients 3 and 4 demonstrated more nuanced clinical interpretations focusing on specific aspects of T-wave/ST segments and QRS complexes respectively. For Coefficient 3, GPT noted that *"A T-wave/ST segment duration of approximately 140.0 ms is within normal limits, suggesting normal ventricular repolarization,"* while for Coefficient 4, it emphasized that *"A QRS duration of approximately 126.0 ms is slightly prolonged, which may indicate a conduction delay such as a bundle branch block."*

Higher-order coefficients (5-8) showed more varied feature focus with less distinctive specialization patterns. For instance, Coefficient 8 was interpreted as focusing on *"normal repolarization timing"* for T-wave/ST segment features, but with less distinct physiological specialization than lower-order coefficients. A detailed breakdown of these GPT-based explanations can be found in Appendix (Section A).

The clinical reasoning provided for each coefficient aligned well with standard ECG interpretation protocols. Coefficient 1's focus on overall ECG morphology parallels the first-glance assessment clinicians make, while Coefficient 2's emphasis on QRS complex corresponds to the subsequent step of examining specific waveform components.

Model	Layer	Feature (ms)	Direction	SHAP Value	Clinical Insight
VPNet (8c)	Raw ECG	T-wave (140.0)	↑	0.35	Normal repolarization
CNN1D	Raw ECG	QRS (112.0)	↑	0.45	Slight prolongation linked to class 0
FCNN	Raw ECG	QRS (120.4)	↓	0.25	Suggests conduction delay
VPNet (8c)	VP Layer	T-wave (142.8)	↑	0.35	Interpretable latent representation
VPNet (4c)	FC Layer	P-wave (5.6)	↑	0.90	Normal atrial activity
Raw ECG	Summed Input	T-wave (142.8)	↑	0.45	Suggests repolarization abnormality

Table 4.1: Representative SHAP-GPT Insights Across Models and Layers

4.4.3 Correlation with SHAP Attribution Patterns

The GPT-generated clinical narratives provided independent validation of our earlier findings regarding coefficient specialization. The clinical interpretation of Coefficient 1 as focusing on overall morphology aligned with our observation that this coefficient showed the highest correlation with Raw ECG. Similarly, the specialized focus of Coefficient 2 on QRS complex features confirmed our earlier finding that this coefficient encoded QRS-specific information.

The graduated importance of the coefficients observed in the FC layer SHAP analysis was reflected in the clarity and consistency of clinical narratives. Lower-order coefficients (1-3) received clear, consistent clinical interpretations, while higher-order coefficients

showed more varied and less distinctive clinical reasoning, confirming their diminished contribution to model decisions.

4.4.4 Implications for Explainable AI in Healthcare

These GPT-generated clinical explanations demonstrate that VPNet’s attribution patterns are not merely statistically significant but clinically meaningful. The clear specialization of VP coefficients enables translation of model decisions into familiar clinical language, potentially increasing trust and adoption among healthcare providers.

The consistency between GPT’s clinical reasoning and standard ECG interpretation suggests that VPNet’s learned representations encode physiologically relevant signal components. This alignment between statistical learning and clinical knowledge provides a promising pathway for explainable AI in healthcare applications, where interpretability is not just a technical requirement but a clinical necessity.

Chapter 5

Conclusion

This thesis introduced a multi-level interpretability framework for ECG classification that combines SHAP-based attribution with GPT-based clinical explanations. The goal was to move beyond conventional black-box models by making their internal decision processes more transparent and clinically meaningful.

We evaluated three neural network architectures, a 1D CNN, a fully connected neural network (FCNN), and a Variable Projection Network (VPNet), using the MIT-BIH Arrhythmia dataset. Our focus was on explaining not just the final outputs of these models, but also their intermediate representations, particularly the coefficient-based encodings in VPNet. This approach enabled a granular understanding of how specific timepoints and waveform components influence model predictions.

Our SHAP analyses showed interesting contrasts across models. CNN1D and FCNN displayed diffuse and inconsistent attribution patterns, with little alignment to known ECG patterns. In contrast, VPNet models demonstrated focused, physiologically coherent attributions, especially in the QRS complex and T-wave regions. Importantly, individual VP coefficients showed clear specialization: some captured global waveform morphology while others focused on specific diagnostic regions like ventricular depolarization or repolarization.

These quantitative findings were further supported by GPT-generated clinical narratives. The textual explanations validated our interpretation of SHAP patterns and mapped the learned features to recognizable clinical terminology. GPT’s explanations consistently reinforced that VPNet’s internal logic aligns well with cardiological reasoning, in contrast to the more uninterpretable CNN and FCNN models.

In addition to interpretability, we also explored the trade-off between compression

and detail. The 4-coefficient VPNet model achieved comparable performance to the 8-coefficient version while concentrating SHAP importance in fewer, more interpretable dimensions. This suggests that lower-complexity models may offer a better balance between performance, interpretability, and computational efficiency.

In conclusion, our results indicate that VPNet offers a promising architecture for explainable AI in ECG analysis. Its Hermite-based projection layer not only improves representation but also enhances interpretability at multiple levels. When combined with SHAP and GPT, VPNet enables end-to-end transparency, from raw input to decision justification, in language clinicians can understand.

This thesis contributes to the growing body of work on interpretable machine learning in healthcare by demonstrating how signal decomposition, attribution methods, and language models can be integrated into a unified framework. While prior studies have focused on either attribution (e.g., SHAP, saliency maps) or language-based reporting separately, this work bridges the two, enabling model reasoning to be both visualized and verbalized in a clinically grounded manner. Future work could extend this approach to multiclass ECG classification, incorporate real-time feedback into clinical workflows, or explore grounding GPT explanations in even more strong biomedical knowledge bases.

Ultimately, this research points toward a path where accuracy and interpretability are not at odds, but synergistically integrated to support trustworthy and effective AI in medicine.

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Appendix A

GPT-Generated Explanations

The following explanations were generated by GPT based on SHAP-derived feature importance for each model and layer.

```
1  [
2  {
3    "model_name": "VPNet (8 coeffs)",
4    "layer_name": "Raw ECG",
5    "class_index": 0,
6    "insights": [
7      {
8        "feature": "T-wave / ST segment (almost equal to 140.0 ms)",
9        "importance": 0.35,
10       "direction": "increases",
11       "rank": 1,
12       "clinical_reasoning": "A T-wave/ST segment duration of approximately 140.0 ms is within the normal range, suggesting normal
13         ↳ ventricular repolarization. This feature increases the likelihood of class 0, indicating a normal or non-pathological
14         ↳ ECG."
15     },
16     {
17       "feature": "T-wave / ST segment (almost equal to 137.2 ms)",
18       "importance": 0.3,
19       "direction": "increases",
20       "rank": 2,
21       "clinical_reasoning": "A T-wave/ST segment duration of approximately 137.2 ms is also within the normal range, reinforcing
22         ↳ the likelihood of normal ventricular repolarization. This feature increases the likelihood of class 0, similar to the
23         ↳ feature ranked 1, but with slightly less influence."
24     },
25     {
26       "feature": "T-wave / ST segment (almost equal to 165.2 ms)",
27       "importance": 0.2,
28       "direction": "decreases",
29       "rank": 3,
30       "clinical_reasoning": "A T-wave/ST segment duration of approximately 165.2 ms is slightly prolonged, which may indicate
31         ↳ early signs of repolarization abnormalities. This feature decreases the likelihood of class 0, suggesting a potential
32         ↳ deviation from normal ECG patterns."
33     },
34     {
35       "feature": "T-wave / ST segment (almost equal to 162.4 ms)",
36       "importance": 0.1,
37       "direction": "decreases",
38       "rank": 4,
39       "clinical_reasoning": "A T-wave/ST segment duration of approximately 162.4 ms is prolonged, potentially indicating
40         ↳ repolarization abnormalities. This feature decreases the likelihood of class 0, aligning with clinical concerns for
41         ↳ prolonged repolarization."
42     }
43   ]
44 }
```

```

35     {
36         "feature": "T-wave / ST segment (almost equal to 168.0 ms)",
37         "importance": 0.05,
38         "direction": "decreases",
39         "rank": 5,
40         "clinical_reasoning": "A T-wave/ST segment duration of approximately 168.0 ms is considered prolonged and may be associated
↪ with repolarization abnormalities. This feature decreases the likelihood of class 0, consistent with clinical
↪ observations of prolonged T-wave/ST segments."
41     }
42 ]
43 },
44 {
45     "model_name": "VPNet (4 coeffs)",
46     "layer_name": "Raw ECG",
47     "class_index": 0,
48     "insights": [
49         {
50             "feature": "T-wave / ST segment (almost equal to 145.6 ms)",
51             "importance": 0.35,
52             "direction": "increases",
53             "rank": 1,
54             "clinical_reasoning": "The T-wave and ST segment duration of approximately 145.6 ms is associated with normal ventricular
↪ repolarization. This duration is within the normal range, suggesting a healthy cardiac function, which increases the
↪ likelihood of classifying the ECG as normal."
55         },
56         {
57             "feature": "T-wave / ST segment (almost equal to 142.8 ms)",
58             "importance": 0.25,
59             "direction": "increases",
60             "rank": 2,
61             "clinical_reasoning": "A T-wave and ST segment duration of approximately 142.8 ms is slightly shorter but still within
↪ normal limits. It supports the classification of the ECG as normal, although its impact is less pronounced than the
↪ feature ranked 1."
62         },
63         {
64             "feature": "T-wave / ST segment (almost equal to 148.4 ms)",
65             "importance": 0.2,
66             "direction": "decreases",
67             "rank": 3,
68             "clinical_reasoning": "A T-wave and ST segment duration of approximately 148.4 ms is slightly longer, which may indicate
↪ early signs of repolarization abnormalities. This feature decreases the likelihood of classifying the ECG as normal."
69         },
70         {
71             "feature": "T-wave / ST segment (almost equal to 140.0 ms)",
72             "importance": 0.15,
73             "direction": "increases",
74             "rank": 4,
75             "clinical_reasoning": "A T-wave and ST segment duration of approximately 140.0 ms is on the lower end of the normal range,
↪ suggesting efficient ventricular repolarization, thus increasing the likelihood of a normal classification."
76         },
77         {
78             "feature": "T-wave / ST segment (almost equal to 151.2 ms)",
79             "importance": 0.05,
80             "direction": "decreases",
81             "rank": 5,
82             "clinical_reasoning": "A T-wave and ST segment duration of approximately 151.2 ms is longer than normal, potentially
↪ indicating delayed repolarization. This feature decreases the likelihood of classifying the ECG as normal."
83         }
84     ]
85 },
86 {
87     "model_name": "CNN1D",
88     "layer_name": "Raw ECG",
89     "class_index": 0,
90     "insights": [
91         {
92             "feature": "QRS complex (almost equal to 112.0 ms)",
93             "importance": 0.45,
94             "direction": "increases",

```

```

95     "rank": 1,
96     "clinical_reasoning": "A QRS duration of approximately 112.0 ms is slightly prolonged, which can be indicative of a
    ↳ conduction delay. This feature increases the likelihood of class 0, possibly due to its association with certain cardiac
    ↳ conditions that the model has learned to recognize."
97 },
98 {
99     "feature": "QRS complex (almost equal to 117.6 ms)",
100     "importance": 0.35,
101     "direction": "increases",
102     "rank": 2,
103     "clinical_reasoning": "A QRS duration of approximately 117.6 ms is more prolonged than 112.0 ms, suggesting a more
    ↳ significant conduction delay. This feature also increases the likelihood of class 0, but its slightly lower importance
    ↳ compared to the 112.0 ms QRS complex may indicate that the model prioritizes the specific pattern or context in which
    ↳ the 112.0 ms duration appears."
104 },
105 {
106     "feature": "QRS complex (almost equal to 109.2 ms)",
107     "importance": 0.15,
108     "direction": "increases",
109     "rank": 3,
110     "clinical_reasoning": "A QRS duration of approximately 109.2 ms is close to the upper limit of normal. It increases the
    ↳ likelihood of class 0, potentially due to its borderline nature, which might be relevant in specific clinical contexts
    ↳ recognized by the model."
111 },
112 {
113     "feature": "QRS complex (almost equal to 106.4 ms)",
114     "importance": 0.05,
115     "direction": "increases",
116     "rank": 4,
117     "clinical_reasoning": "A QRS duration of approximately 106.4 ms is within normal limits but on the higher end. It slightly
    ↳ increases the likelihood of class 0, possibly due to its association with early signs of conduction issues."
118 },
119 {
120     "feature": "QRS complex (almost equal to 100.8 ms)",
121     "importance": 0.02,
122     "direction": "decreases",
123     "rank": 5,
124     "clinical_reasoning": "A QRS duration of approximately 100.8 ms is considered normal and decreases the likelihood of class
    ↳ 0. This feature suggests that the model associates normal QRS durations with a lower probability of class 0, possibly
    ↳ due to the absence of conduction delays."
125 }
126 ]
127 },
128 {
129     "model_name": "FCNN",
130     "layer_name": "Raw ECG",
131     "class_index": 0,
132     "insights": [
133         {
134             "feature": "T-wave / ST segment (almost equal to 140.0 ms)",
135             "importance": 0.35,
136             "direction": "increases",
137             "rank": 1,
138             "clinical_reasoning": "A T-wave/ST segment duration of approximately 140.0 ms is considered within normal limits. This
            ↳ feature increases the likelihood of class 0, suggesting a normal or non-pathological ECG finding. The T-wave and ST
            ↳ segment are crucial for assessing repolarization, and normal durations typically indicate healthy cardiac function."
139         },
140         {
141             "feature": "QRS complex (almost equal to 120.4 ms)",
142             "importance": 0.25,
143             "direction": "decreases",
144             "rank": 2,
145             "clinical_reasoning": "A QRS complex duration of approximately 120.4 ms is slightly prolonged, which can indicate a
            ↳ conduction delay such as a bundle branch block. This feature decreases the likelihood of class 0, suggesting a potential
            ↳ abnormality. The QRS complex duration is critical for diagnosing ventricular conduction issues."
146         },
147         {
148             "feature": "T-wave / ST segment (almost equal to 162.4 ms)",
149             "importance": 0.15,

```

```

150     "direction": "decreases",
151     "rank": 3,
152     "clinical_reasoning": "A T-wave/ST segment duration of approximately 162.4 ms is prolonged, which may indicate
↪ repolarization abnormalities. This feature decreases the likelihood of class 0, suggesting a potential pathological
↪ finding. Prolonged repolarization can be associated with conditions like ischemia or electrolyte imbalances."
153 },
154 {
155     "feature": "QRS complex (almost equal to 114.8 ms)",
156     "importance": 0.15,
157     "direction": "increases",
158     "rank": 4,
159     "clinical_reasoning": "A QRS complex duration of approximately 114.8 ms is at the upper limit of normal. This feature
↪ increases the likelihood of class 0, indicating a non-pathological finding. While slightly prolonged, it is not
↪ typically associated with significant conduction abnormalities."
160 },
161 {
162     "feature": "T-wave / ST segment (almost equal to 145.6 ms)",
163     "importance": 0.1,
164     "direction": "increases",
165     "rank": 5,
166     "clinical_reasoning": "A T-wave/ST segment duration of approximately 145.6 ms is slightly prolonged but still within a range
↪ that can be considered normal for some individuals. This feature increases the likelihood of class 0, suggesting a
↪ benign finding. Slight prolongation without other abnormalities is often not clinically significant."
167 }
168 ]
169 },
170 {
171     "model_name": "Raw ECG",
172     "layer_name": "Summed Input",
173     "class_index": 0,
174     "insights": [
175         {
176             "feature": "T-wave / ST segment (almost equal to 142.8 ms)",
177             "importance": 0.45,
178             "direction": "increases",
179             "rank": 1,
180             "clinical_reasoning": "A prolonged T-wave/ST segment can indicate repolarization abnormalities, which are often associated
↪ with ischemic heart conditions. This feature's increase suggests a higher likelihood of such conditions."
181         },
182         {
183             "feature": "T-wave / ST segment (almost equal to 140.0 ms)",
184             "importance": 0.35,
185             "direction": "increases",
186             "rank": 2,
187             "clinical_reasoning": "Similar to the top feature, a prolonged T-wave/ST segment is indicative of potential ischemic
↪ changes. However, its slightly shorter duration compared to the top feature may suggest a less pronounced abnormality."
188         },
189         {
190             "feature": "T-wave / ST segment (almost equal to 137.2 ms)",
191             "importance": 0.15,
192             "direction": "decreases",
193             "rank": 3,
194             "clinical_reasoning": "A shorter T-wave/ST segment duration might be associated with normal repolarization, thus decreasing
↪ the likelihood of ischemic conditions."
195         },
196         {
197             "feature": "T-wave / ST segment (almost equal to 145.6 ms)",
198             "importance": 0.05,
199             "direction": "increases",
200             "rank": 4,
201             "clinical_reasoning": "An even more prolonged T-wave/ST segment could indicate severe repolarization issues, but its lower
↪ importance suggests it is less influential in this decision context."
202         },
203         {
204             "feature": "QRS complex (almost equal to 134.4 ms)",
205             "importance": 0.02,
206             "direction": "decreases",
207             "rank": 5,
208             "clinical_reasoning": "A prolonged QRS complex can be associated with bundle branch blocks or ventricular hypertrophy.
↪ However, its minimal importance and decreasing direction suggest it plays a minor role in this decision."

```

```

209     }
210   ]
211 },
212 {
213   "model_name": "VPNet FC Layer (8c)",
214   "layer_name": "First FC Layer",
215   "class_index": 0,
216   "insights": [
217     {
218       "feature": "P-wave region (almost equal to 0.0 ms)",
219       "importance": 0.9,
220       "direction": "increases",
221       "rank": 1,
222       "clinical_reasoning": "The P-wave represents atrial depolarization. A P-wave region close to 0.0 ms suggests a very short or
↪ absent P-wave, which may indicate atrial fibrillation or other atrial abnormalities. This feature significantly
↪ increases the likelihood of class 0, possibly indicating an arrhythmia."
223     },
224     {
225       "feature": "P-wave region (almost equal to 19.6 ms)",
226       "importance": 0.7,
227       "direction": "decreases",
228       "rank": 2,
229       "clinical_reasoning": "A P-wave region of 19.6 ms is within the normal range for atrial depolarization duration. This
↪ feature decreases the likelihood of class 0, suggesting normal atrial activity, which contrasts with the feature ranked
↪ 1."
230     },
231     {
232       "feature": "P-wave region (almost equal to 2.8 ms)",
233       "importance": 0.5,
234       "direction": "increases",
235       "rank": 3,
236       "clinical_reasoning": "A P-wave region of 2.8 ms is unusually short, potentially indicating atrial conduction issues. This
↪ feature increases the likelihood of class 0, supporting the presence of atrial abnormalities."
237     },
238     {
239       "feature": "P-wave region (almost equal to 5.6 ms)",
240       "importance": 0.3,
241       "direction": "increases",
242       "rank": 4,
243       "clinical_reasoning": "A P-wave region of 5.6 ms is shorter than normal, which may suggest atrial conduction delay or
↪ abnormality, increasing the likelihood of class 0."
244     },
245     {
246       "feature": "P-wave region (almost equal to 11.2 ms)",
247       "importance": 0.1,
248       "direction": "decreases",
249       "rank": 5,
250       "clinical_reasoning": "A P-wave region of 11.2 ms is within the lower normal range, decreasing the likelihood of class 0 and
↪ suggesting normal atrial conduction."
251     }
252   ]
253 },
254 {
255   "model_name": "VPNet FC Layer (4c)",
256   "layer_name": "First FC Layer",
257   "class_index": 0,
258   "insights": [
259     {
260       "feature": "P-wave region (almost equal to 5.6 ms)",
261       "importance": 0.9,
262       "direction": "increases",
263       "rank": 1,
264       "clinical_reasoning": "The P-wave represents atrial depolarization. A P-wave duration of 5.6 ms is within the normal range,
↪ suggesting normal atrial conduction. This feature increases the likelihood of class 0, potentially indicating a normal
↪ sinus rhythm."
265     },
266     {
267       "feature": "P-wave region (almost equal to 2.8 ms)",
268       "importance": 0.7,

```

```

269     "direction": "decreases",
270     "rank": 2,
271     "clinical_reasoning": "A P-wave duration of 2.8 ms is shorter than the normal range, which may indicate an abnormal atrial
    ↳ conduction or ectopic atrial rhythm. This feature decreases the likelihood of class 0, suggesting a deviation from
    ↳ normal sinus rhythm."
272 },
273 {
274     "feature": "P-wave region (almost equal to 0.0 ms)",
275     "importance": 0.5,
276     "direction": "decreases",
277     "rank": 3,
278     "clinical_reasoning": "A P-wave duration of 0.0 ms suggests the absence of a P-wave, which could indicate atrial
    ↳ fibrillation or junctional rhythm. This feature decreases the likelihood of class 0, indicating a significant deviation
    ↳ from normal sinus rhythm."
279 },
280 {
281     "feature": "P-wave region (almost equal to 8.4 ms)",
282     "importance": 0.3,
283     "direction": "increases",
284     "rank": 4,
285     "clinical_reasoning": "A P-wave duration of 8.4 ms is slightly longer than the normal range but may still be considered
    ↳ within normal limits depending on the context. This feature slightly increases the likelihood of class 0, potentially
    ↳ indicating a normal or slightly prolonged atrial conduction."
286 }
287 ]
288 },
289 {
290     "model_name": "VP Layer (8c) - Coeff 1",
291     "layer_name": "VP Layer",
292     "class_index": 0,
293     "insights": [
294         {
295             "feature": "T-wave / ST segment (almost equal to 142.8 ms)",
296             "importance": 0.35,
297             "direction": "increases",
298             "rank": 1,
299             "clinical_reasoning": "A T-wave / ST segment duration of approximately 142.8 ms is indicative of a normal repolarization
    ↳ phase, which is generally associated with a healthy cardiac function. This feature increases the likelihood of class 0,
    ↳ suggesting normal cardiac activity."
300         },
301         {
302             "feature": "T-wave / ST segment (almost equal to 140.0 ms)",
303             "importance": 0.25,
304             "direction": "increases",
305             "rank": 2,
306             "clinical_reasoning": "A T-wave / ST segment duration of approximately 140.0 ms also suggests normal cardiac repolarization.
    ↳ However, its slightly shorter duration compared to 142.8 ms may indicate a marginally faster repolarization, which still
    ↳ supports normal cardiac function but with slightly less influence on the model's decision for class 0."
307         },
308         {
309             "feature": "T-wave / ST segment (almost equal to 137.2 ms)",
310             "importance": 0.15,
311             "direction": "decreases",
312             "rank": 3,
313             "clinical_reasoning": "A T-wave / ST segment duration of approximately 137.2 ms is on the shorter side and may suggest a
    ↳ faster repolarization phase, which could be associated with certain cardiac conditions. This feature decreases the
    ↳ likelihood of class 0, indicating a potential deviation from normal cardiac function."
314         },
315         {
316             "feature": "T-wave / ST segment (almost equal to 145.6 ms)",
317             "importance": 0.15,
318             "direction": "increases",
319             "rank": 4,
320             "clinical_reasoning": "A T-wave / ST segment duration of approximately 145.6 ms is slightly longer and may indicate a slower
    ↳ repolarization phase, which is generally considered normal but could be associated with certain conditions if
    ↳ significantly prolonged. This feature increases the likelihood of class 0, supporting normal cardiac function."
321         },
322         {
323             "feature": "QRS complex (almost equal to 134.4 ms)",

```

```

324     "importance": 0.1,
325     "direction": "decreases",
326     "rank": 5,
327     "clinical_reasoning": "A QRS complex duration of approximately 134.4 ms is longer than the typical range (80-120 ms) and may
    ↪ suggest ventricular conduction delay. This feature decreases the likelihood of class 0, indicating a potential
    ↪ abnormality in cardiac conduction."
328   }
329 ]
330 },
331 {
332   "model_name": "VP Layer (4c) - Coeff 1",
333   "layer_name": "VP Layer",
334   "class_index": 0,
335   "insights": [
336     {
337       "feature": "T-wave / ST segment (almost equal to 142.8 ms)",
338       "importance": 0.35,
339       "direction": "increases",
340       "rank": 1,
341       "clinical_reasoning": "A T-wave / ST segment duration of approximately 142.8 ms is associated with normal ventricular
    ↪ repolarization. This feature increases the likelihood of class 0, suggesting normal cardiac function."
342     },
343     {
344       "feature": "T-wave / ST segment (almost equal to 140.0 ms)",
345       "importance": 0.25,
346       "direction": "increases",
347       "rank": 2,
348       "clinical_reasoning": "A T-wave / ST segment duration of approximately 140.0 ms is also indicative of normal ventricular
    ↪ repolarization, but with slightly less influence than the 142.8 ms duration. This feature also increases the likelihood
    ↪ of class 0, supporting normal cardiac function."
349     },
350     {
351       "feature": "T-wave / ST segment (almost equal to 137.2 ms)",
352       "importance": 0.15,
353       "direction": "decreases",
354       "rank": 3,
355       "clinical_reasoning": "A T-wave / ST segment duration of approximately 137.2 ms may suggest a slightly faster repolarization
    ↪ process, which could be less typical. This feature decreases the likelihood of class 0, indicating a potential deviation
    ↪ from normal repolarization."
356     },
357     {
358       "feature": "T-wave / ST segment (almost equal to 145.6 ms)",
359       "importance": 0.15,
360       "direction": "increases",
361       "rank": 4,
362       "clinical_reasoning": "A T-wave / ST segment duration of approximately 145.6 ms is within the normal range and increases the
    ↪ likelihood of class 0, indicating normal cardiac function."
363     },
364     {
365       "feature": "QRS complex (almost equal to 134.4 ms)",
366       "importance": 0.1,
367       "direction": "decreases",
368       "rank": 5,
369       "clinical_reasoning": "A QRS complex duration of approximately 134.4 ms is longer than typical, which may suggest a
    ↪ conduction delay. This feature decreases the likelihood of class 0, indicating a potential abnormality in ventricular
    ↪ depolarization."
370     }
371   ]
372 }
373 ]

```

Code A.1: Full output of GPT explanations

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